Wireless Questions

13 questions | Original English questions and source answers

Question 1

Which 802.11 frame type is association response?

Options

A. protected frame

B. action

C. control

D. management

Answer

D. management

Source explanation

There are three main types of 802.11 frames: the Data Frame, the Management Frame and the Control Frame. Association
Response belongs to Management Frame. Association response is sent in response to an association request.

Source: CCNA Training » Wireless Quiz.pdf | Question 1

Question 2

Which statement about Link Aggregation when implemented on a Cisco Wireless LAN Controller is true?

Options

A. When enabled the WLC bandwidth drops to 500 Mbps

B. The EtherChannel must be configured in “mode active”

C. One functional physical port is needed to pass client traffic

D. To pass client traffic two or more ports must be configured

Answer

C. One functional physical port is needed to pass client traffic

Source explanation

Link aggregation (LAG) is a partial implementation of the 802.3ad port aggregation standard. It bundles all of the
controller’s distribution system ports into a single 802.3ad port channel.
Restriction for Link aggregation:
+ LAG requires the EtherChannel to be configured for ‘mode on’ on both the controller and the Catalyst switch -> Answer
'The EtherChannel must be configured in “mode active”' is not correct.
+ If the recommended load-balancing method cannot be configured on the Catalyst switch, then configure the LAG
connection as a single member link or disable LAG on the controller -> Answer 'To pass client traffic two or more ports
must be configured' is not correct while answer 'One functional physical port is needed to pass client traffic' is correct.
Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-5/configuration-guide/b_cg75/b_cg75_chapter_0100010.html

Source: CCNA Training » Wireless Quiz.pdf | Question 2

Question 3

Which mode allows access points to be managed by Cisco Wireless LAN Controllers?

Options

A. lightweight

B. mobility express

C. bridge

D. autonomous

Answer

A. lightweight

Source explanation

A Lightweight Access Point (LAP) is an AP that is designed to be connected to a wireless LAN (WLAN) controller (WLC).
APs are “lightweight,” which means that they cannot act independently of a wireless LAN controller (WLC). The WLC
manages the AP configurations and firmware. The APs are “zero touch” deployed, and individual configuration of APs is
not necessary.
Reference: https://www.cisco.com/c/en/us/support/docs/wireless/aironet-1200-series/70278-lap-faq.html

Source: CCNA Training » Wireless Quiz.pdf | Question 3

Question 4

Several new coverage cells are required to improve the Wi-Fi network of an organization. Which two standard designs are
recommended? (Choose two)

Options

A. 5GHz channel selection requires an autonomous access point.

B. Adjacent cells with overlapping channels use a repeater access point.

C. 5GHz provides increased network capacity with up to 23 nonoverlapping channels.

D. Cells that overlap one another are configured to use nonoverlapping channels.

E. For maximum throughput, the WLC is configured to dynamically set adjacent access points to the same channel.

Answer

C. 5GHz provides increased network capacity with up to 23 nonoverlapping channels.

D. Cells that overlap one another are configured to use nonoverlapping channels.

Source explanation

The 802.11 standard defines 23 20MHz wide channels in the 5 GHz spectrum. Each channel is spaced 20MHz apart and
separated into three Unlicensed National Information Infrastructure (UNII) bands.
Reference:
https://documentation.meraki.com/MR/WiFi_Basics_and_Best_Practices/Channel_Planning_Best_Practices

Source: CCNA Training » Wireless Quiz.pdf | Question 4

Question 5

Which QoS Profile is selected in the GUI when configuring a voice over WLAN deployment?

Options

A. Gold

B. Silver

C. Platinum

D. Bronze

Answer

C. Platinum

Source explanation

Cisco Unified Wireless Network solution WLANs support four levels of QoS: Platinum/Voice, Gold/Video, Silver/Best
Effort (default), and Bronze/Background.
Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-4/configuration/guides/consolidated/b_cg74_CONSOLIDATED/b_cg74_CONSOLIDATED_chapter_01010111.html

Source: CCNA Training » Wireless Quiz.pdf | Question 5

Question 6

Which type of wireless encryption is used for WPA2 in pre-shared key mode?

Options

A. AES-256

B. RC4

C. TKIP with RC4

D. AES-128

Answer

D. AES-128

Source explanation

WPA 2 implements the National Institute of Standards and Technology (NIST)-recommended Advanced Encryption
Standard (AES) encryption algorithm with the use of Counter Mode with Cipher Block Chaining Message Authentication
Code Protocol (CCMP). AES Counter Mode is a block cipher that encrypts 128-bit blocks of data at a time with a 128-bit
encryption key.
Reference:
https://www.cisco.com/c/en/us/support/docs/wireless-mobility/wireless-lan-wlan/67134-wpa2-config.html
Also here is a quick comparison between WPA, WPA2 & WPA3:
Reference: https://www.networkacademy.io/ccna/wireless/client-authentication-methods

	WEP	WPA	WPA2	WPA3
Released	1997	2003	2004	2018
Encryption protocol	RC4	TKIP	CCMP (with AES)	GCMP (with AES)
Session Key	40 or 104bits	128bits	128bits	128/256bits
Authentication	Open system (OSA)/ Shared Key (SRA)	802.1X / Pre-shared Key	802.1X / Pre-shared Key	SAE (Simultaneous Authentication of Equals)
Level of security	Low	Medium	High	Very High
Vulnarabilities	Easily cracked due to weak RC4	Vulnerable to TKIP exploits	PSK vulnerable to brute force	Improved against DDoS, Brute Force and MITM attacks

Source: CCNA Training » Wireless Quiz.pdf | Question 6

Question 7

Which two values or settings must be entered when configuring a new WLAN in the Cisco Wireless LAN Controller GUI?
(Choose two)

Options

A. Profile name

B. QoS settings

C. ip address of one or more access points

D. management interface settings

E. SSID

Answer

A. Profile name

E. SSID

Source: CCNA Training » Wireless Quiz.pdf | Question 7

Question 8

Which feature on the Cisco Wireless LAN Controller when enabled restricts management access from specific networks?

Options

A. Flex ACL

B. CPU ACL

C. RADIUS

D. TACACS

Answer

B. CPU ACL

Source explanation

Whenever you want to control which devices can talk to the main CPU, a CPU ACL is used.
Note: CPU ACLs only filter traffic towards the CPU, and not any traffic exiting or generated by the CPU.
Reference:
https://www.cisco.com/c/en/us/support/docs/wireless/4400-series-wireless-lan-controllers/109669-secure-wlc.html

Source: CCNA Training » Wireless Quiz.pdf | Question 8

Question 9

What is a benefit of using a Cisco Wireless LAN Controller?

Options

A. Central AP management requires more complex configurations

B. It eliminates the need to configure each access point individually

C. Unique SSIDs cannot use the same authentication method

D. It supports autonomous and lightweight APs

Answer

B. It eliminates the need to configure each access point individually

Source: CCNA Training » Wireless Quiz.pdf | Question 9

Question 10

What is a design element when deploying an 802.11b wireless infrastructure?

Options

A. setting the maximum data rate to 54 Mbps on the Cisco Wireless LAN Controller

B. disabling TPC so that access points can negotiate signal levels with their attached wireless devices.

C. configuring access points to provide clients with a maximum of 5 Mbps

D. allocating non overlapping channels to access points that are in close physical proximity to one another

Answer

D. allocating non overlapping channels to access points that are in close physical proximity to one another

Source: CCNA Training » Wireless Quiz.pdf | Question 10

Question 11

When configuring a WLAN with WPA2 PSK in the Cisco Wireless LAN Controller GUI, which two formats are available
to select? (Choose two)

Options

A. binary

B. base64

C. hexadecimal

D. decimal

E. ASCII

Answer

C. hexadecimal

E. ASCII

Source explanation

When configuring a WLAN with WPA2 Preshared Key (PSK), we can choose the encryption key format as either ASCII or
HEX.
Reference:
https://www.cisco.com/c/en/us/td/docs/wireless/controller/9800/config-guide/b_wl_16_10_cg/multi-preshared-key.pdf

Source: CCNA Training » Wireless Quiz.pdf | Question 11

Question 12

Which unified access point mode continues to serve wireless clients after losing connectivity to the Cisco Wireless LAN
Controller?

Options

A. flex connect

B. mesh

C. local

D. sniffer

Answer

A. flex connect

Source explanation

In previous releases, whenever a FlexConnect access point disassociates from a controller, it moves to the standalone mode.
The clients that are centrally switched are disassociated. However, the FlexConnect access point continues to serve locally
switched clients. When the FlexConnect access point rejoins the controller (or a standby controller), all clients are
disconnected and are authenticated again. This functionality has been enhanced and the connection between the clients and
the FlexConnect access points are maintained intact and the clients experience seamless connectivity. When both the access
point and the controller have the same configuration, the connection between the clients and APs is maintained.
Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/7-4/configuration/guides/consolidated/b_cg74_CONSOLIDATED/b_cg74_CONSOLIDATED_chapter_010001101.html

Source: CCNA Training » Wireless Quiz.pdf | Question 12

Question 13

An engineer must configure a WLAN using the strongest encryption type for WPA2-PSK. Which cipher fulfills the
configuration requirement?

Options

A. AES

B. RC4

C. WEP

D. TKIP

Answer

A. AES

Source explanation

Many routers provide WPA2-PSK (TKIP), WPA2-PSK (AES), and WPA2-PSK (TKIP/AES) as options. TKIP is actually an older encryption protocol introduced with WPA to replace the very-insecure WEP encryption at the time. TKIP is
actually quite similar to WEP encryption. TKIP is no longer considered secure, and is now deprecated. In other words, you
shouldn’t be using it.
AES is a more secure encryption protocol introduced with WPA2 and it is currently the strongest encryption type for
WPA2-PSK.

Source: CCNA Training » Wireless Quiz.pdf | Question 13